



Discovering Knowledge

AI-Powered Diabetic Retinopathy Detection and Grading System

Final Year Project Proposal

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1. Introduction

Diabetic Retinopathy (DR) is a progressive retinal disorder caused by long-term diabetes and is among the leading causes of preventable blindness worldwide. Early detection through retinal fundus imaging significantly reduces the risk of permanent vision loss. However, manual grading of retinal images requires trained ophthalmologists, is time consuming, and becomes impractical for large-scale screening in resource-constrained environments.

Recent advancements in Artificial Intelligence (AI), particularly Convolutional Neural Networks (CNNs), have demonstrated high potential in automated medical image classification. Despite promising results, many existing DR detection systems suffer from methodological limitations such as improper dataset splitting leading to data leakage, severe class imbalance, insufficient use of clinically relevant evaluation metrics, and limited explainability. This project proposes a structured, patient-level validated, imbalance-aware, explainable deep learning framework for multi-class DR grading with deployment feasibility analysis.

2. Objective

To design and implement a patient-level validated, imbalance-aware, and explainable deep learning system for multi-class Diabetic Retinopathy grading (severity levels 0–4) and deploy it through a web-based diagnostic interface.

3. Problem Description

Automated Diabetic Retinopathy grading systems typically focus on improving classification accuracy while overlooking critical clinical and methodological considerations.

One major issue is data leakage, which occurs when left and right eye images of the same patient are split across training and testing sets. This results in artificially inflated performance metrics and poor real-world generalization.

Additionally, DR datasets are highly imbalanced. Early-stage cases dominate publicly available datasets, whereas advanced stages (which are clinically critical) are underrepresented. Models trained without addressing imbalance tend to misclassify severe cases, which poses a significant medical risk.

Most studies rely primarily on accuracy as a performance metric, which is insufficient for medical diagnosis tasks. Medical AI systems require evaluation through metrics such as Precision, Recall, F1-score, and Quadratic Weighted Kappa to ensure clinically meaningful reliability.

Furthermore, many models lack explainability. Without visual justification (e.g., highlighting lesions), clinicians may not trust AI-assisted predictions.

Therefore, there is a need for a reliable DR grading framework that:

- Uses patient-level dataset splitting
- Addresses class imbalance
- Evaluates performance using medical-grade metrics
- Provides visual explanations
- Demonstrates deployment feasibility

4. Literature Review

4.1 Deep Learning in Diabetic Retinopathy Detection

Recent advancements in deep learning have significantly improved automated Diabetic Retinopathy (DR) detection using retinal fundus images. Gulshan et al. [1] demonstrated that convolutional neural networks trained on large-scale datasets such as EyePACS can achieve performance comparable to ophthalmologists in detecting referable DR. Similarly, Abràmoff et al. [2] developed an autonomous AI-based diagnostic system that achieved high clinical reliability.

Modern architectures such as ResNet [3] and EfficientNet [4] have further enhanced image classification performance through deeper network design and transfer learning. These architectures are widely adopted in medical imaging due to their strong feature extraction capabilities. However, many existing studies primarily focus on improving accuracy while overlooking issues such as improper dataset splitting, class imbalance, and real-world deployment constraints.

4.2 Explainable AI in Medical Imaging

Despite high predictive performance, deep learning models are often considered “black-box” systems. In medical applications, interpretability is essential for clinical trust and transparency. Grad-CAM, introduced by Selvaraju et al. [5], provides visual explanations by highlighting image regions that influence model predictions.

Although many DR classification studies include Grad-CAM visualizations, explainability is often treated as an additional visualization step rather than an integrated evaluation component. Limited attention has been given to systematically combining explainability with medical-grade performance evaluation and deployment considerations.

4.3 Research Gap

Based on existing literature, the following gaps are identified:

- Limited emphasis on patient-level dataset splitting to prevent data leakage
- Insufficient handling of severe class imbalance
- Over-reliance on accuracy instead of medical-grade metrics such as Quadratic Weighted Kappa
- Minimal integration of explainability into structured evaluation

- Limited focus on deployment feasibility and inference efficiency

The proposed project aims to address these gaps by developing a patient-level validated, imbalance-aware, explainable, and deployment-oriented multi-class DR grading framework.

5. Methodology

5.1 Dataset Characteristics

The EyePACS dataset used in this project consists of 35,126 high-resolution retinal fundus images collected from approximately 17,563 patients, with each patient having left and right eye images. The images are labeled into five Diabetic Retinopathy severity classes (0–4). The dataset is highly imbalanced, with 73.5% of images belonging to Class 0 (No DR), while advanced stages (Class 3 and 4) together represent less than 5% of the data. Due to the high image resolution (ranging from approximately 2560×1920 to 4752×3168), all images will be resized to 224×224 for computational efficiency. A patient-level split (70/15/15) will be performed to prevent data leakage and ensure reliable evaluation.

5.2 Data Splitting Strategy

- Patient-level split (to prevent data leakage)
- 70% Training
- 15% Validation
- 15% Testing

5.3 Preprocessing

- Image resizing to 224×224
- Pixel normalization
- Data augmentation (horizontal flip, slight rotation, brightness adjustment)
- Augmentation applied only to training data

5.4 Class Imbalance Handling

- Balanced sampling for training set
- Optional class weighting in cross-entropy loss
- Validation and testing sets retain natural distribution

5.5 Model Architecture

Two CNN architectures will be implemented:

1. ResNet18 (Baseline Model)
2. EfficientNet-B0 (Primary Model)

Both models will use transfer learning with modified final layers for five-class softmax output.

5.6 Training Strategy

- Cross-entropy loss
- Adam optimizer
- Learning rate scheduling
- Early stopping
- Hyperparameter tuning

5.7 Performance Evaluation Metrics

- Accuracy
- Precision, Recall, F1-score (per class)
- Confusion Matrix
- Quadratic Weighted Kappa
- Inference Time
- Model Size Comparison

5.8 Explainability

Grad-CAM will be implemented to visualize image regions influencing model predictions. Heatmaps will be overlaid on retinal fundus images to provide interpretability analysis.

5.9 System Architecture

System Workflow

1. User uploads retinal image
2. Image preprocessing
3. CNN model inference
4. Softmax prediction (Grade 0–4)
5. Grad-CAM visualization generation
6. Display of:
 - Predicted Grade
 - Confidence Score
 - Heatmap Overlay
 - Screening Recommendation

5.10 Software Process Model

The Incremental Development Model will be followed:

Phase 1 – Dataset Preparation

Phase 2 – Baseline Model

Phase 3 – Improved Model

Phase 4 – Explainability Integration Phase

5 – Deployment

This ensures continuous testing and refinement.

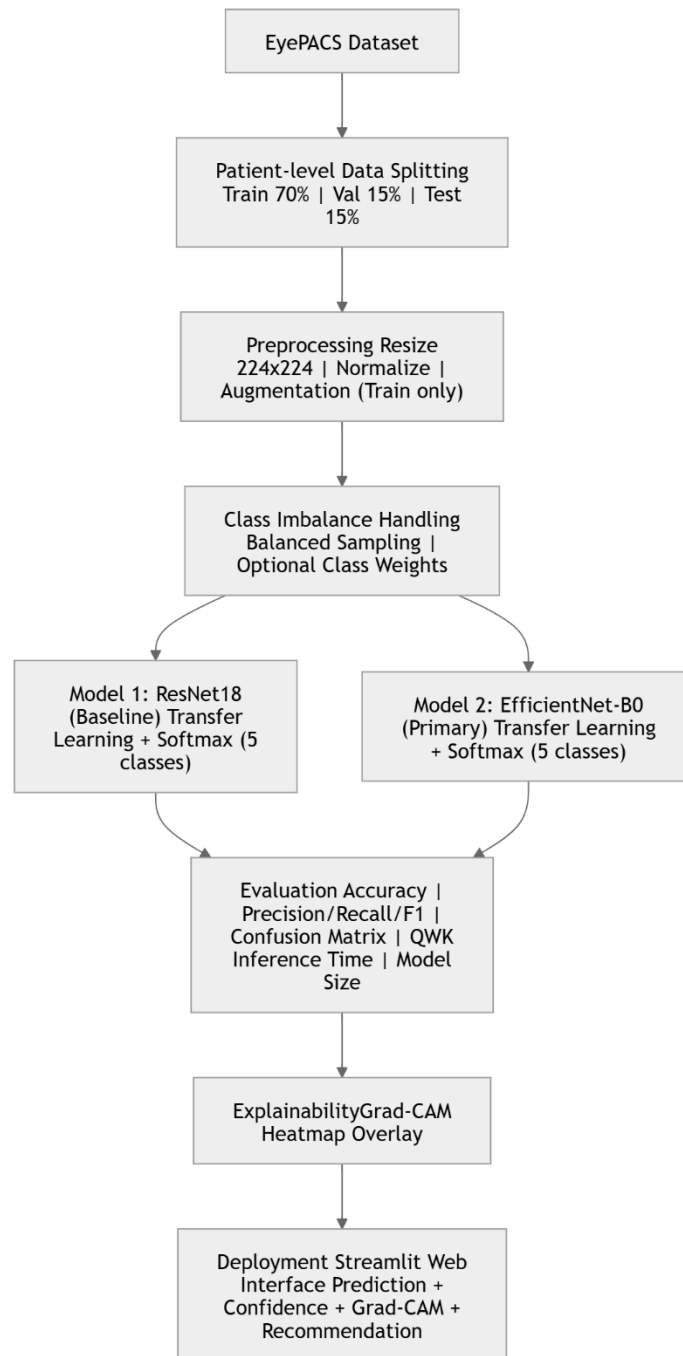


Figure 1 - Methodology

6. Project Scope

6.1 Included

- Multi-class DR grading (0–4)
- Patient-level dataset splitting
- Imbalance-aware training
- Two-model comparison
- Explainability using Grad-CAM
- Web-based deployment
- Medical-grade evaluation metrics

6.2 Not Included

- Real hospital data integration
- Regulatory approval or certification
- Clinical trials
- Hospital information system integration
- Large-scale distributed cloud deployment

7. Feasibility Study

7.1 Risks Involved

Risk	Mitigation
GPU limitations	Use lightweight architectures
Class imbalance	Balanced sampling & class weighting
Overfitting	Early stopping & augmentation
Time constraints	Limit to two core architectures
Data leakage	Strict patient-level splitting

7.2 Resource Requirements

- Google Colab / Kaggle GPU
- Python, PyTorch
- OpenCV, scikit-learn
- 16GB RAM laptop for deployment demo

The project is feasible within the academic timeline.

8. Solution Application Areas

- Diabetic screening centers
- Telemedicine platforms
- Rural healthcare units
- AI-assisted ophthalmology research

The system can assist in early screening, reduce specialist workload, and improve accessibility in underdeveloped regions.

9. Tools and Technologies

- Python
- PyTorch
- OpenCV
- scikit-learn
- NumPy
- Matplotlib
- GitHub
- Google Colab / Kaggle

10. Expertise of the Team Members

Team members have prior experience in:

- CNN-based image classification
- Transfer learning
- Python programming
- Model evaluation techniques

Relevant coursework completed includes:

- Machine Learning
- Artificial Intelligence
- Computer Vision
- Data Structures
- Software Engineering

The project aligns with the academic background and interests of both team members.

11. Milestones

Time Period	Planned Activities
Month 1	Literature review & dataset preparation
Month 2	Patient-level splitting and preprocessing pipeline development
Month 3	Baseline model implementation (ResNet18)
Month 4	Primary model implementation (EfficientNet-B0)
Month 5	Model training, hyperparameter tuning, and evaluation
Month 6	Class imbalance experiments and performance comparison
Month 7	Explainability integration (Grad-CAM) and qualitative analysis

Month 8	Web-based deployment (Streamlit) and system integration
Month 9	Final evaluation and documentation

Gantt Chart:

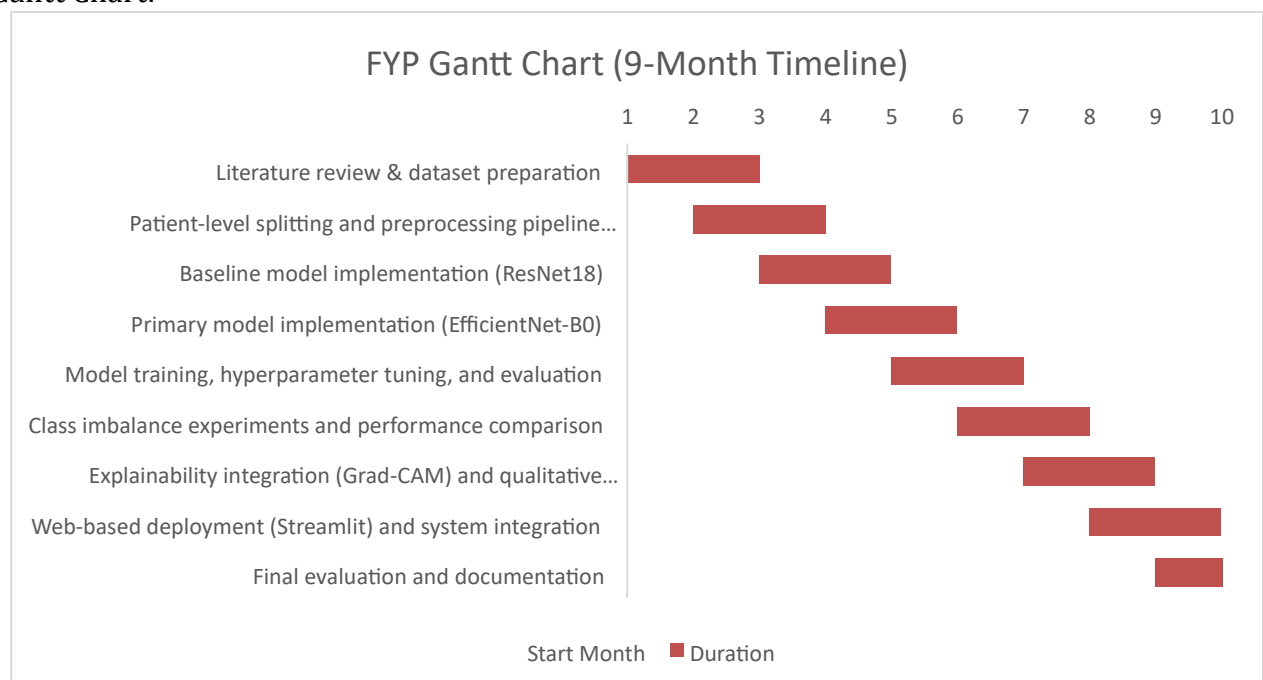


Figure 2 - Gantt Chart

12. References

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- [6] EyePACS Dataset Documentation.